

MALAQUITE BEACH, TX, USA

WMO: 997366

Lat: 27.581N

Lon: 97.216W

Elev: 33

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 37.3	(c) 41.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 32.9	(k) 54.1	(l) 29.4	(m) 53.6	(n) 18.0	(o) 350	(p) 0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.7	(c) 89.0	(d) N/A	(e) 86.8	(f) N/A	(g) 85.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 12.7	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 29.3	(b) 26.8	(c) 24.8	(e) 34.9	(f) 91.4	(g) 3.9	(h) 4.1	(i) 32.0	(j) 94.4	(k) 29.8	(l) 96.8	(m) 27.6	(n) 99.1	(o) 24.7	(p) 102.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	56.4	59.6	64.0	71.2	76.7	81.7	82.3	83.1	81.7	76.5	67.4	60.3
	DBStd	10.86	7.07	8.61	6.12	4.36	3.26	1.88	1.95	1.77	2.90	5.40	7.07	7.19
	HDD50	66	29	19	5	0	0	0	0	0	0	0	3	10
	HDD65	785	270	178	86	10	1	0	0	0	0	4	58	178
	CDD50	8019	227	287	438	636	829	950	1000	1026	951	822	525	328
	CDD65	3262	4	25	55	195	364	500	535	561	501	361	130	31
	CDH74	31585	2	268	10	379	2550	5639	6326	6911	5925	3239	329	7
	CDH80	8718	0	143	2	25	129	1444	1955	2498	2003	516	3	0
(14) Wind	WSAvg	14.2	13.1	13.5	13.7	15.6	16.5	15.7	14.5	12.7	13.8	14.6	13.7	13.2
(15) Precipitation	PrecAvg	32.6	1.6	1.7	2.3	2.1	3.5	2.7	2.4	2.2	5.3	3.7	2.2	1.6
	PrecMax	48.1	5.6	4.7	5.0	9.1	12.3	8.7	13.9	6.7	14.8	16.4	7.1	7.0
	PrecMin	17.9	0.0	0.1	0.1	0.2	0.1	0.0	0.0	0.1	0.9	0.0	0.0	0.2
	PrecStd	8.5	1.5	1.5	1.6	2.1	3.2	2.1	3.3	2.0	3.1	3.7	2.0	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	91.7	75.0	83.9	84.3	89.6	90.1	90.2	90.1	86.9	79.3	74.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	67.9	86.0	73.0	78.3	82.3	86.3	86.8	87.9	87.3	84.5	78.1	72.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	66.6	68.2	71.5	77.1	81.3	84.7	85.4	86.3	86.0	83.7	77.3	71.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	65.4	66.8	70.4	76.0	80.4	84.0	84.8	85.4	85.2	82.7	76.3	69.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	7.8	5.8	5.0	3.7	3.2	3.4	3.7	5.4	6.5	8.2	9.2
		MCDBR	9.3	8.7	6.3	7.8	5.2	5.8	4.7	5.6	6.7	5.7	8.0	9.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.364	0.370	0.389	0.432	0.470	0.447	0.464	0.451	0.440	0.394	0.379	0.355
	taud		2.479	2.463	2.406	2.290	2.215	2.344	2.314	2.349	2.342	2.467	2.463	2.517
	Ebn at Noon		279	286	286	276	264	268	264	267	267	275	271	277
	Edh at Noon		30	33	37	43	46	40	41	40	39	32	30	28
(44) All-Sky Solar Radiation	RadAvg		954	1138	1451	1733	1942	2132	2102	2009	1657	1448	1085	857
	RadStd		103	187	158	165	131	105	166	108	138	144	112	116

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+185

Nomenclature: See separate page